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SECTION 1: LANDSLIDE RISK ASSESSMENT

2.1 Introduction

Landslides risks are pervasive in hilly and mountain landscapes of the globe, and are typical occurrences in Rwanda. The term landslide denotes the downhill movement of slope forming materials under the influence of gravity (Cruden and Varnes, 1996). Landslides are one of the most widespread and effective agents shaping the Earth's surface (Egholm et al., 2013; Wang et al., 2020). With the development and urbanisation of hilly and mountain terrains around the globe, landslide occurrence also frequently intersects with human activities and the built environment, often with disastrous consequences (Sidle and Ochiai, 2006; Lu and Godt, 2013; Froude and Petley, 2018; Haque et al., 2019). While landslides are pervasive Earth surface processes naturally occurring in hilly/mountain landscapes, human activities (e.g., roads, reservoir construction, deforestation, urbanisation, etc.) can also influence their occurrence, extent and timing (Sidle and Ochiai, 2006; Lacroix et al., 2020). Landslide characteristics reflect the very diverse geologic, topographic, environmental, and climatic conditions in which they can occur, resulting in a large diversity of landslide types and processes (Lu and Godt, 2013; Hungr et al., 2014).

The UN's Sendai Framework for Disaster Risk Reduction identifies four priorities for action in order to substantially reduce global landslide disaster risk and related fatalities (UNDRR, 2015). A cornerstone among these priorities is understanding landslide hazard, which describes the likelihood of a landslide of a given magnitude to occur in time and space (Guzzetti et al., 1999). In other words, landslide hazard is ideally characterized by statements of 'what', 'where', 'how strong' and 'how frequent' (Glade et al., 2006).

Authorities and decision makers need maps depicting the areas that may be affected by landslides so that they are considered in development plans and/or that appropriate risk mitigation measures are implemented. A wide variety of methods for assessing landslide susceptibility, hazard and risk are available (Galde et al., 2006; Corominas et al., 2014; Reichenbach et al., 2018).

In this study, the assessment of landslide risk was achieved in five steps (Figure 1). This approach relied on conventional methods that are recommended for local-scale landslide risk assessment (Corominas et al., 2014). The landslide inventory (Step 1) was the key step of this approach. The outcomes of the risk assessment is influenced not only by the quality of the inventory but also by the types of landslides and the abundance of their occurrence.